STA2016 Final Project

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Abstract

This study investigates the relationship between vehicular road infrastructure and nitrogen oxide (NOx) concentrations in Ontario to better understand the spatial and environmental impacts of transportation. Various spatial modeling approaches were employed to assess spatial autocorrelation and identify significant predictors of NOx levels. Results reveal that road infrastructure, particularly local streets and highways, is strongly associated with NOx concentrations. This work underscores the critical role of urban infrastructure in air pollution and suggests future directions for continued research in the domain.

Introduction

With increasing concerns about the impact of human activities on the climate, environment, and public health, this study seeks to contribute to the understanding of these complex interactions by focusing on a specific facet of the issue. Among the various factors influencing environmental degradation, this paper examines the relationship between vehicle usage and nitrogen oxides (NOx) concentration in Ontario.

NOx is produced primarily through the combustion processes in vehicles, power plants, and industrial activities. These compounds are well-documented for their adverse effects on human health, including an increased risk of respiratory illnesses such as asthma and chronic obstructive pulmonary disease (COPD). Additionally, NOx plays a significant role in environmental degradation, serving as a precursor to acid rain, which can have far-reaching consequences for ecosystems and infrastructure. By exploring the association between vehicular emissions and NOx levels, this research aims to shed light on the environmental and health implications of transportation-related pollution in Ontario.

According to a 2010 study by the government of Canada, roughly 55% of the Nitrogen Oxides Emissions were from road transportation, as shown in Figure 1.

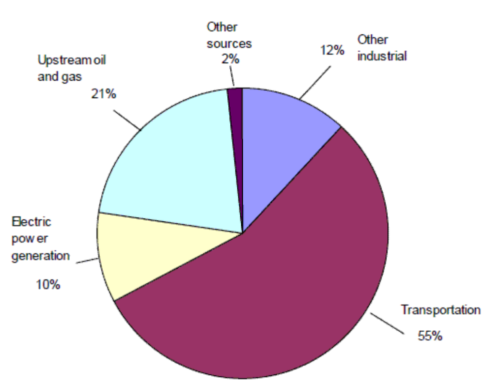

Figure 1: 2010 Canadian NOx Emissions by Source (total 2.06 Mt)

Methods

Data Sources

Air pollution levels

The data for Nitrogen Oxide levels were collected from the Ontario Ministry of the Environment, Conservation, and Parks. The government branch collects hourly geostatistical data for air pollutants across 36 different locations within Ontario. A webscraping script was written in python to collect the hourly data from each of the 36 stations separate web pages. Additionally, a second script was written to collect station information including geographical position, elevation, and height of the air intake. Below in Table 1 is a summary of the features collected.

Table 1: NOx Concentration Data

Station details

Column	Description
Station_name	Location of Station
Stations_ID	Unique Identifier
address	street address of station
lon	Longitude
lat	Lattitude
Elevation	Elevation at the location of the station
Height_of_Intake	Height above ground of air intake

Station pollutant measurements

Column	Description
Station_ID	Unique Identifier
Pollutant	For this study only used NOx
Date	Date of recording
H1	NOx levels (ppb) at hour 1
...	...
H24	NOx levels (ppb) at hour 24

Ontario Road Infrastructure

A public database maintained by the Ontario Ministry of Natural Resources and Forestry containing over 250,000 km of Ontario roads. This dataset will act as a proxy for vehicle usage. The database includes the road classification, number of lanes, road surface, speed limit, and more. This database is available as a shapefile for download from the ministry's website. Due to the file's size and the computational limitations of the computer used for this project, the entire dataset cannot be displayed as a map. Below in Table 2 is a full list of all the features provided by the dataset.

Table 2: Ontario Road Infrastructure dataset

Column	Description
ROAD_NET_E	Unique Identifier
FULL_STREE	Full name of street
L_FIRST_HO	left Lowest house number within segment
L_LAST_HOU	Left Highest house number within segment
L_HOUSE_NU	Left House number structure
L_STANDARD	Municipality of homes on Left
R_FIRST_HO	Right Lowest house number within segment
R_LAST_HOU	Right Highest house number within segment
R_HOUSE_NU	Right House number structure
R_STANDARD	Municipality of homes on Right
ALT_STREET	Alternate street name
EFFECTIVE_	Date of recording
SPEED_LIMI	Speed limit
NUMBER_OF_	Number of lanes
PAVEMENT_S	Status of road surface (paved/ unpaved)

ROAD_CLASS	Road classification (e.g. local, highway)
JURISDICTI	Governing jurisdiction
ROUTE_NUMB	Route for highways
DIRECTION_	Direction of traffic flow (one way or both)
GEOMETRY_U	Date of latest geometry update
SHIELD_TYP	Highway shield type
ROAD_ELEME	Road element type (e.g., road, ferry)
LENGTH	Length of the segment (m)
geometry	Shapefile Geometry points

Data Processing

To perform the intended statistical analysis the different datasets were loaded into R, processed, and then merged. To simplify the NOx dataset, the data was aggregated. First, a boxplot was done to understand the NOx trend throughout the day, the resulting graph is shown in Figure 2, note outlier points have been removed for readability.

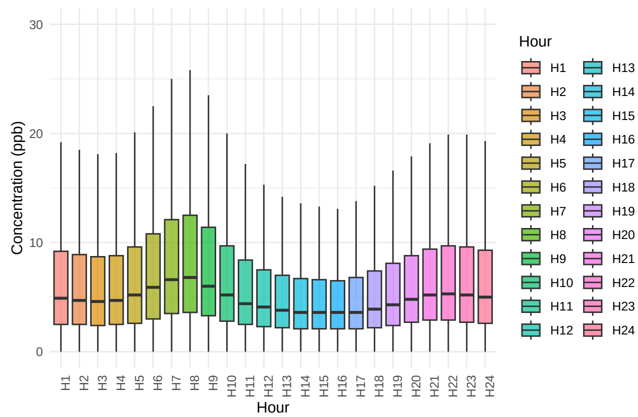

Figure 2: Average hourly NOx levels

Based on the plot, we can summarize that the variance within a day is minimal. Next, the yearly trend of a few key locations was examined. Figure 3 illustrates the average NOx ppb over the course of 2023 for three different stations.

While the data exhibits variation throughout the year, there is no discernible trend observed across the locations. Given the absence of a clear temporal pattern, it is statistically justifiable to calculate the average of the daily statistics for each station as a representative measure of NOx concentration at each location.

After collecting each stations average NOx reading, the distribution of NOx levels was analyzed both in its untransformed state and after applying a logarithmic transformation. Figure 4 illustrates the distribution for both cases. The log-transformed data more closely approximates a normal distribution, suggesting its suitability for subsequent analysis. As a result, the log-transformed NOx levels will be used in this study.

Using the log-transformed data, a spatial visualization of the monitoring stations and their corresponding log-transformed NOx levels is presented in Figure 5. A visual inspection of the map reveals that the highest NOx levels are concentrated in larger urban centers, such as Toronto and Windsor. Additionally, the station density is noticeably higher within these metropolitan areas.

Given that the Thunder Bay station is geographically isolated from the other locations, it will be excluded from the dataset. This exclusion ensures a higher spatial density of stations within the primary study domain, facilitating a more robust analysis.

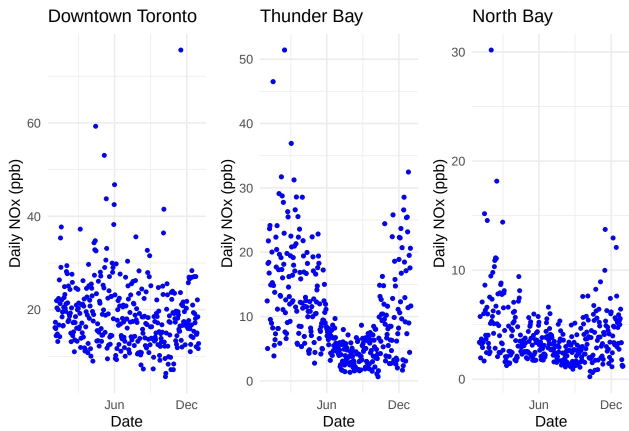

Figure 3: Daily NOx levels across stations in 2023

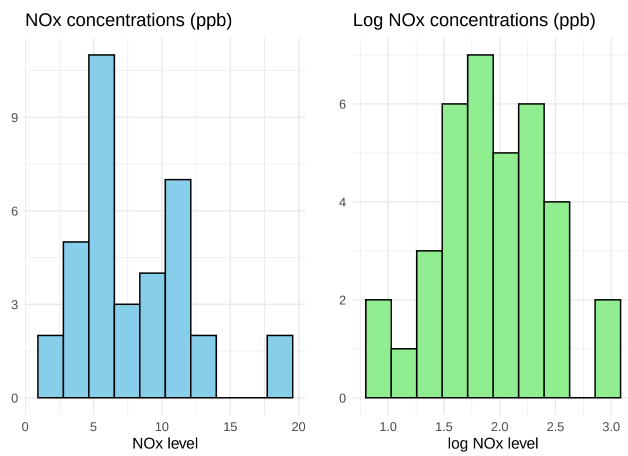

Figure 4: NOx yearly concentations under no transformation and log transformation

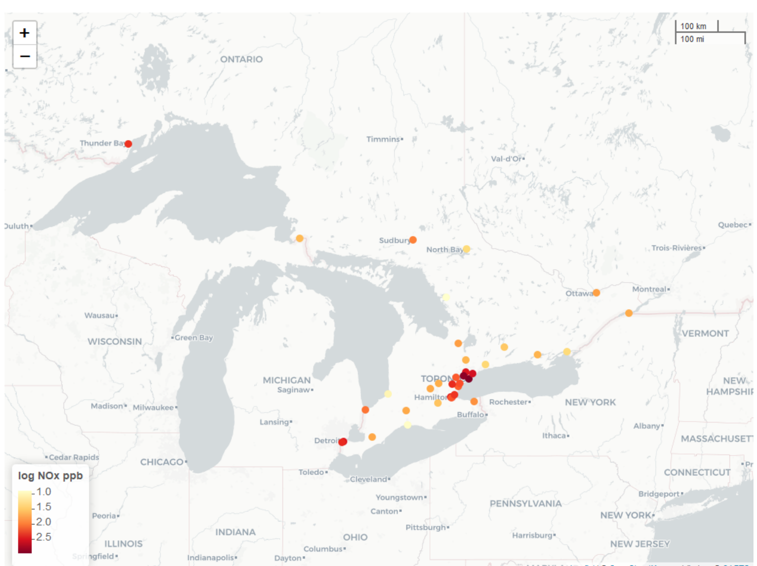

Figure 5: Map of NOx stations with average yearly concentration readings

Spatial Computations

To facilitate further spatial analysis and integration of the datasets, a spatial relationship between the monitoring stations and road network information was established. Road network data were incorporated as a covariate in the spatial analysis for each station. This was achieved by creating a 50 km buffer around each monitoring station and then performing a spatial intersection of all the roads within each buffer, with the roads split based on road type. The relative proportion of each road type within the entire dataset, based on total road length, is summarized in Table 3. This categorization provides a detailed representation of the road network's composition in the vicinity of each station and supports subsequent analyses of the relationship between road infrastructure and NOx levels. To improve modeling accuracy, the data was scaled to between 0 and 1.

Table 3: Road Category Distribution

Category	Percentage
Local / Street	52%
Resource / Recreation	21%
Local / Strata	10%
Collector	8%
Arterial	6%
Expressway / Highway	3%

Note that since the road infrastructure dataset often breaks a single road into smaller segments, and that each segment can vary wildly in length, it was determined that a simple count of road observations within each buffer would not be a sufficient metric. So instead of the count, the length of each road segment within a buffer was calculated. While this was far more computationally expensive, it was only required to be done once to capture the necessary data. On the left, Figure 6 depicts the 50km buffers around the stations and on the right is the Belleville station's buffer with all the intersecting roads within it.

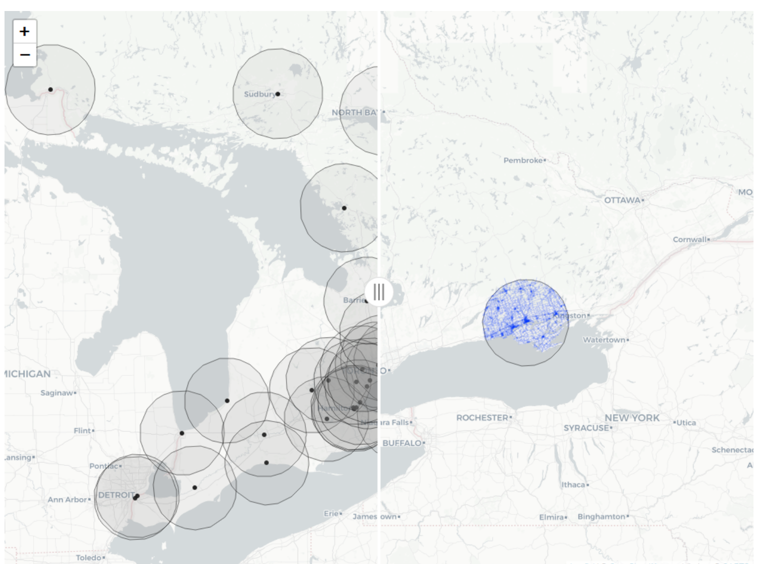

Figure 6: Map of buffers and raods within Belleville buffer

Data Exploration

Variograms

The first step in the data exploration process after performing a review of the maps was to fit a semivariogram to the data to understand the spatial autocorrelation between the stations. Below Figure 7 shows the full cloud variogram on the left and on the right the binned variogram, with the binning range and number of bins determined through trail and error.

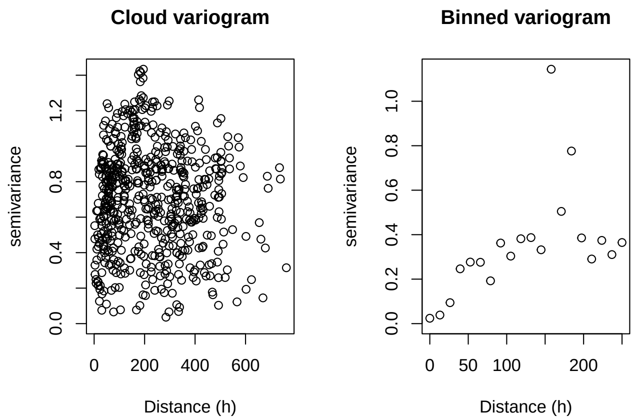

Figure 7: Semivariogram for log NOx concentrations

Based on the plots we see that there is a definite trend within the semivariograms. By analyzing the binned semivariogram we can estimate the parameters of the semivariogram should be:

- **sill**: ~0.4
- **range:**~ 50- 200
- **nugget:** ~0

The theoretical semivariogram was fitted to the data using a range of candidate models. A subset of these fitted semivariograms is presented in Figure 8. While there is no universally strict computational criterion for selecting the optimal variogram across all methods (e.g., AIC is only applicable for ML and REML approaches), the final selection was made based on visual inspection and alignment with the observed empirical semivariogram.

From the evaluated models, Table 4 shows the fitted parameters and results. The exponential variogram fitted using ordinary least squares (OLS) was determined to provide the best representation of the data. This model will be used in subsequent analyses requiring a variogram. The exponential variogram is mathematically defined as:

$$\gamma(h) = \tau^2 + \sigma^2(1 - \exp(-h/\phi))$$

For the fitted variogram the final parameters were:

- **nugget** (tau squared): 0
- **sill** (sigma squared): 0.50
- **range** (phi): 69.00

Which are all within the ranges of the expected values described from visual inspection of the variograms above

Table 4: Results of fitting theoretical variograms

Method	Model	Nugget	Range	Sill	Smoothness	SSE	AIC	BIC
OLS	exponential	0.0000000	69.00167	0.4961765	0.5	0.7317450	0.00000	0.00000
OLS	gaussian	0.0615057	82.77481	0.3574655	0.5	0.7070393	0.00000	0.00000
WLS	exponential	0.0205717	178.22937	0.8884236	0.5	54.5875794	0.00000	0.00000
WLS	gaussian	0.0234070	67.33883	0.5228022	0.5	51.7648178	0.00000	0.00000
ML	exponential	0.0043515	33.90936	0.1854275	0.5	0.0000000	37.42173	43.64312
ML	gaussian	0.0148063	25.37189	0.1603437	0.5	0.0000000	35.82462	42.04601
RML	exponential	0.0048522	37.05483	0.1961204	0.5	0.0000000	36.41581	42.63721
RML	gaussian	0.0149958	25.65812	0.1671293	0.5	0.0000000	35.21742	41.43881

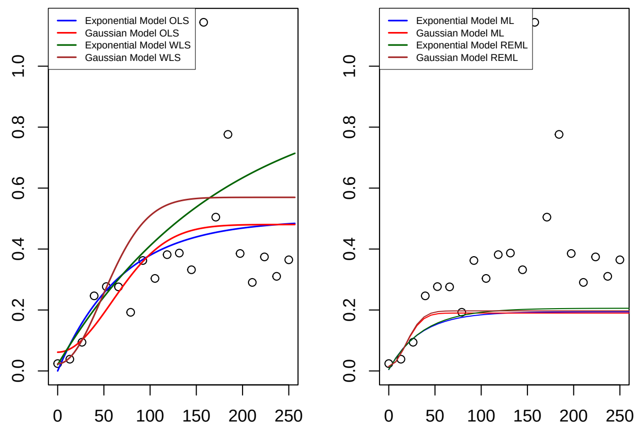

Figure 8: Fitted theoretical variograms

Moran's I

The spatial autocorrelation of the NOx monitoring stations were assessed using Moran's I. Moran's I is a measure that evaluates the degree of spatial clustering, dispersion, or randomness of a spatial process. Mathematically, we define Moran's I as:

$$I = \frac{1}{s^2} \frac{\sum_i \sum_j w_{ij}(y_i - \overline{y})(y_j - \overline{y})}{\sum_i \sum_j w_{ij}}$$

where,

$$s^2 = \frac{\sum_i (y_i - \overline{y})^2}{n} \quad \text{and} \quad \overline{y} = \sum_{i=1}^{n} \frac{y_i}{n}$$

and w_{ij} is the spatial weighting between point i and j. To conduct the Moran's I test, it is necessary to define a spatial weights matrix that describes the relationships between the stations. Given that the data is not areal, conventional weight matrices such as the queen or rook contiguity are not applicable. Instead, two alternative approaches—K-Nearest Neighbors (KNN) and inverse distance weighting—were employed.

- **KNN**: For each station, an equal weight is assigned to the k nearest stations. The value of k was determined to be 4 after conducting a hyperparameter search aimed at minimizing the p-value.
- **Inverse distance**: The weight between two stations is defined inversely proportional to the distance separating them, meaning that stations located closer to each other are assigned higher weights.

Under both weight matrix configurations, the null hypothesis of no spatial autocorrelation was strongly rejected, with p-values of 6.05e-6 for KNN and 6.44e-5 for inverse distance. Corresponding Moran's I values were 0.35 and 0.18, respectively, indicate the presence of significant positive spatial autocorrelation among the stations. Figure 9 shows the results of the monte carlo simulations, once again showing that under both weight matrices the NOx concentration are positively spatially auto correlated. As the K-nearest neighbour has the lower p-value and higher moran's I score we will use that moving forward.

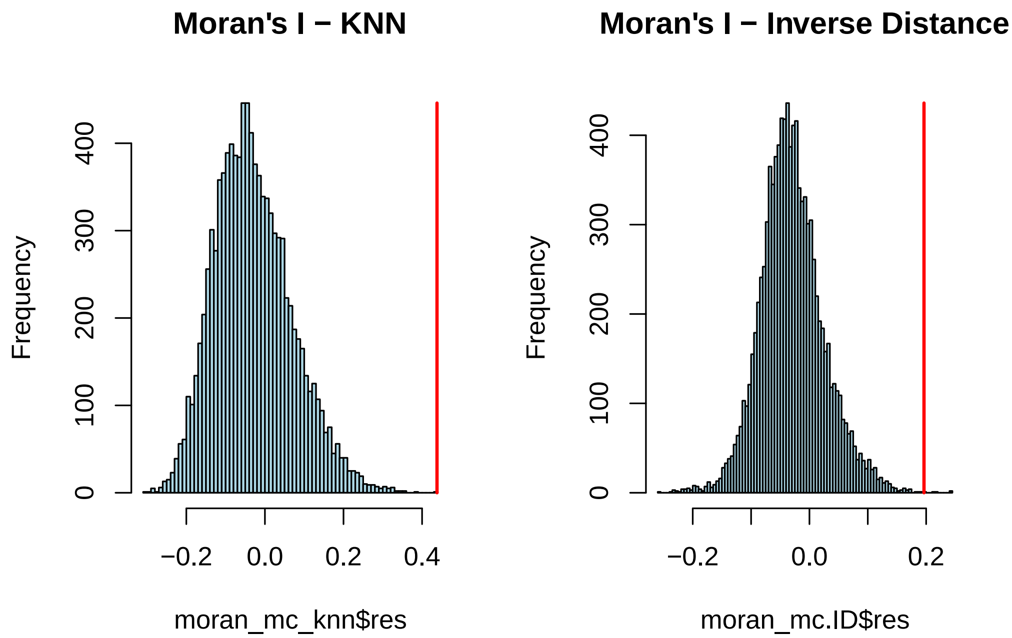

Figure 9: Moran's I MC simulation under different weight matrices

Spatial Modeling

Now that we have determined a semivariogram and weight matrix we look to create spatial models of our process to better understand the underlying relationships. Multiple potential models were investigated, with the summary of results shown in Table 5, with the GAM model with spatial covariates having the best AIC while a Moran's I test of the residuals of all the models showed no spatial correlation, implying that they have fit the process well.

Table 5: Model performance

			Residual's Moran's I	
Model	Covariates	AIC	Statistic	P-value
Simple Kriging	FALSE	-	-0.026	0.48
Ordinary Kriging	FALSE	-	-0.030	0.51
SAR	FALSE	46.5	0.055	0.20
SAR	TRUE	42.15	-0.027	0.49
CAR	FALSE	43.49	-0.010	0.41
CAR	TRUE	42.18	-0.010	0.35
GAM	FALSE	43.92	-0.030	0.50
GAM	TRUE	39.67	-0.070	0.67

Additionally Figure 10 shows the residual of the SAR, CAR models and the predictions of the GAM models (with covariates) within Ontario. Based on the SAR and CAR residuals we can see that they are not overly spatially correlated, with potentially slightly lower residuals in the Toronto area. The GAM predictions show the Toronto area as the highest area, with lower values in more rural areas. Which is in line with what is expected.

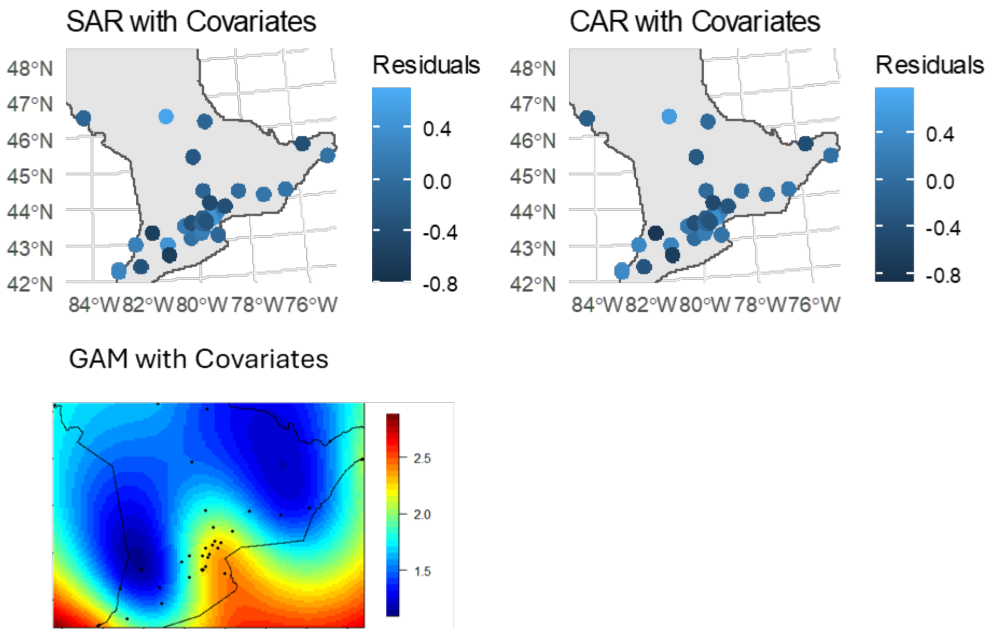

Figure 10: Prediction and residual outputs of models

Kriging

There are 3 types of Kriging. Simple, ordinary, and universal kriging. All three types are spatial interpolation methods to predict an unknown value of a variable at unobserved locations. The key difference is in the assumptions of the three forms of kriging, where:

- **Simple**: Assumes the mean of the process (this case NOx concentrations) is known and constant across the entire area
- **Ordinary:** Assumes that the mean is constant but unknown, allowing for a more flexible model
- **Universal:** Assumes the mean follows a trend in the x and y and other spatial covariates

First simple and ordinary kriging were performed and the Moran's I of the residuals was calculated. Following that a linear and quadratic model was fitted to the spatial data, however as there were no significant spatial predictors it was decided not to fit a universal kriging model. Figure 11 shows the resulting predictions and standard errors of the kriging processes. In both cases Toronto has the highest log NOx concentrations and the errors are low near observed locations and increase in regions with no near observations.

SAR

The simultaneous autorgressive (SAR) models incorporate spatial dependence directly into the dependent variable. In general a SAR model can be defined as:

$$Y = X\beta + \rho WY + \epsilon$$

where Y is the dependent variable W is the spatial weights matrix (here we will use the k-nearest neighbours

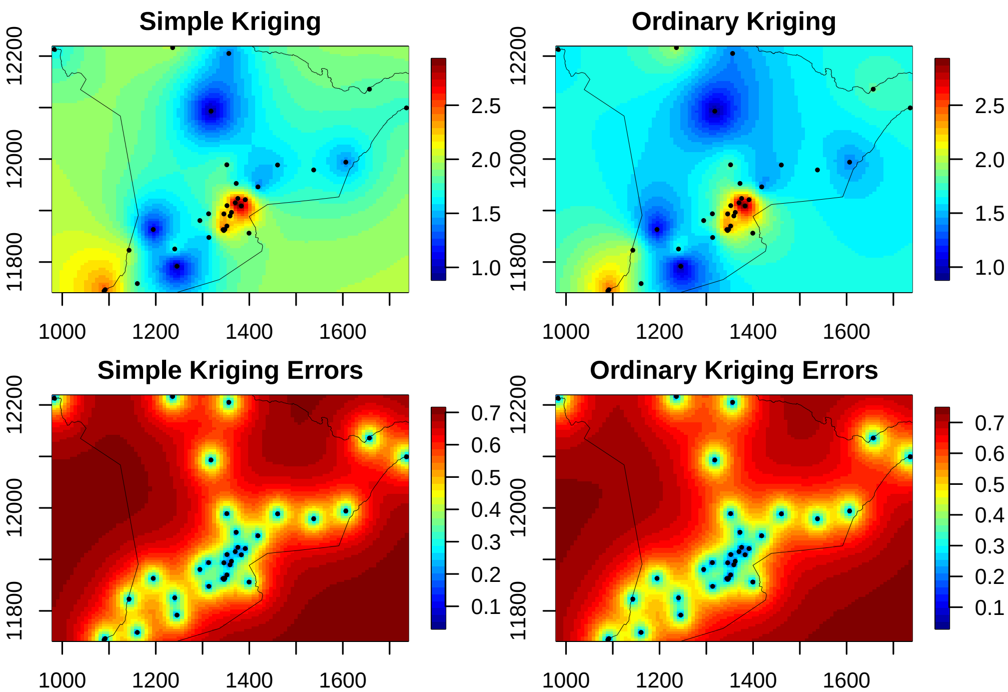

Figure 11: Prediction and standard errors of simple and ordinary kriging

defined as above), ρ is the spatial autogressive parameter, X is the matrix of covariates, β is a vector of regression coefficients, and ϵ is the error term

There are 3 kinds of SAR models:

- **Spatial Lag:** For this model we assume that the error term ϵ is iid so we arrive at an equation of

$$Y = (I - \rho W)^{-1} X\beta + (I - \rho W)^{-1}\epsilon$$

- **Spatial Error:** For this model we no longer assume the errors are iid and are instead defined as $\epsilon = \lambda W\epsilon + \nu$, so we arrive at our new definition:

$$Y = X\beta + (I - \lambda W)^{-1}\nu$$

- **Spatial Lag-Error:** This model essentially combines the 2 above, allowing for spatial autocorrelation in both the dependent variable and in the error terms

In this paper we will use the Spatial Lag-Error model as it is the most robust of the 3. A model was fitted first without any covariates and then with covariates to understand the effect adding the covariates have on the model. The parameters of the SAR model with covariates were:

$$\lambda = -0.42 \quad \text{and} \quad \rho = 0.41$$

Table 6 shows the coefficients and the statistical significance of each. Based on this we can summarize that the two most statistically important road classes are the local streets and the express/ highways and that they are both positively correlated to the NOx concentrations.

Table 6: SAR Lag-error coefficients

Term	Estimate	Std. Error	z value	Pr(greater than z)
Intercept	-2.68485	1.74489	-1.5387	0.1238800
Local Street	6.36913	2.84092	2.2419	0.0249700
Resource Recreation	-3.44394	1.88619	-1.8259	0.0678700
Arterial	1.48390	0.93518	1.5867	0.1125700
Local Strata	0.28318	1.25914	0.2249	0.8220600
Collector	0.34602	0.47914	0.7222	0.4701900
Expressway Highway	3.00639	0.68028	4.4194	0.0000099

CAR

The conditional autoregressive (CAR) model assumes a conditional dependence structure (instead of a joint distribution like in the SAR models). With each station's NOx concentration conditionally dependent on it's neighbours. Again we created the model with and without the covariates to compare the results. The parameters for the CAR model with covariates was:

$$\lambda = 0.002$$

Since λ is small, we can conclude the spatial structure captured by the weights matrix W has minimal influence on the NOx concentrations. Instead, the NOx concentrations are more strongly explained by the covariates (road types). When adding in the covariates the only statistically significant covariate is the expressway/ highway density which is positively correlated to the NOx concentrations.

Table 7: CAR coefficients

Term	Estimate	Std. Error	z value	Pr(greater than z)
Intercept	-1.42476	1.99694	-0.7135	0.4755551
Local Street	5.37453	3.23504	1.6613	0.0966433
Resource Recreation	-1.88824	2.25654	-0.8368	0.4027137
Arterial	0.97359	1.03875	0.9373	0.3486186
Local Strata	-0.45670	1.44172	-0.3168	0.7514163
Collector	0.17347	0.57903	0.2996	0.7644962
Expressway Highway	2.74229	0.78311	3.5018	0.0004621

GAM

The Generalized Additive Model (GAM) is used to model spatial data by allowing non-linear relationships between the dependent variable and predictor variables while accounting for spatial autocorrelation. GAMs incorporate smooth functions of spatial coordinates. As such the response variable is assumed to be smooth and an additive function of the predictors. These models are flexible and allow for complex, non-linear spatial relationships without assuming a strict parametric form. A sweep of different possible k-values was performed. Again the model was first built without the covariates and then again with the covariates. There were no statistically significant covariates for this model.

Table 8: GAM coefficients

Term	Estimate	Std. Error	t value	Pr(greater than t)
Intercept	1.9500000	0.4189000	4.655	0.000172
Local Street	-0.0002164	0.0003296	-0.656	0.519471
Resource Recreation	0.0000254	0.0002018	0.126	0.901014

Arterial	-0.0002391	0.0004822	-0.496	0.625642
Local Strata	-0.0011860	0.0006525	-1.817	0.084970
Collector	0.0007152	0.0004433	1.613	0.123148
Expressway Highway	0.0008267	0.0017290	0.478	0.638105

Conclusions

Across both the SAR and CAR models adding in the covariates improved the AIC of the models and reduced the spatial correlation of the residuals. However, interestingly with the GAM model while the AIC improved when including the covariates, the spatial correlation of the residuals also increased. This may suggest that the GAM model was overfitting.

What this result suggests is that the quantity of road infrastructure is correlated to the NOx levels within a region. While this is an expected result, it is interesting to note that the most statistically significant predictors across the different models were the local streets (SAR) and the highways (SAR and CAR). Additionally, in the SAR model the recreation road class was negatively correlated to the NOx levels (though the p-value was just over the traditional cutoff of 5%).

Limitations and Extensions

This work was in part limited by the computational power of the available computer. A more powerful computer would have allowed for more experimental tests (e.g., different buffer sizes around the stations). Additionally, the NOx readings were limited to the existing stations which were concentrated near Toronto, a possible extension of this work could include building more monitoring stations to collect more readings in under represented areas currently. Other extensions of this work could include layering in more data including traffic volumes (not publicly available at the time of writing).